



MÁSTER EN FINANZAS CUANTITATIVAS



GUÍA DIDÁCTICA Módulo 10

CURSO 2025/2026

Módulo 10: Teoría del Valor Extremo

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Coordinador:

Manuel Sánchez Sánchez Profesor del Departamento de Teoría Económica y Economía Matemática
UNED. Fac. de Ciencias Económicas y Empresariales.



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PRESENTACION

El Máster en Finanzas Cuantitativa se integra por 10 módulos que se agrupan en los siguientes tres cursos de experto.

Curso de Experto en Métodos Computacionales en Finanzas

Módulo 1: Programación en Python y en R

Módulo 2: Modelos de Simulación

Módulo 3: Datos Financieros y Aprendizaje Automático

Curso de Experto en Métodos Econométricos Aplicados en Finanzas

Módulo 4: Modelos de Series Temporales

Módulo 5: Modelos de Volatilidad y Correlación

Módulo 6: Modelos de No linealidad

Curso de Experto en Análisis Financiero y Gestión de Riesgos

Módulo 7: Valoración de Instrumentos Financieros

Módulo 8: Gestión de Carteras de Inversión

Módulo 9: Gestión de Riesgos Financieros

Módulo 10: Teoría del Valor Extremo

Atención directa al alumno:

Prof. Manuel Sánchez Sánchez

UNED

Facultad de Ciencias Económicas y Empresariales

Departamento de Teoría Económica y Economía Matemática

Pº. Senda del Rey, 11

28040 Madrid

Teléfonos: 913 986399 – Lunes y Miércoles (mañanas) -

E.mail: mjsanchez@cee.uned.es

Extreme Value Theory

Extreme Value Theory: An Introduction

- Conventional statistical models are based upon estimates of the average or typical behaviour of a series
- Such models often perform poorly when the focus switches to extreme events
- In particular, statistics based on a normal distribution will systematically under-estimate the probability of extreme events in most financial time series as they cannot capture the fat tails
- Extreme value distributions are sufficiently flexible that they can capture these fat tails much more accurately

Extreme Value Theory: An Introduction

- For example, a normal distribution would suggest the probability of a return of -10.84% or lower in a 30-year period on A-rated corporate bonds is 0.00008%, whereas based on an extreme value distribution it is 1.4% (source: Levine, D. (2009) Modelling tail behavior with extreme value theory Risk Management Society of Actuaries 17, 15-18).

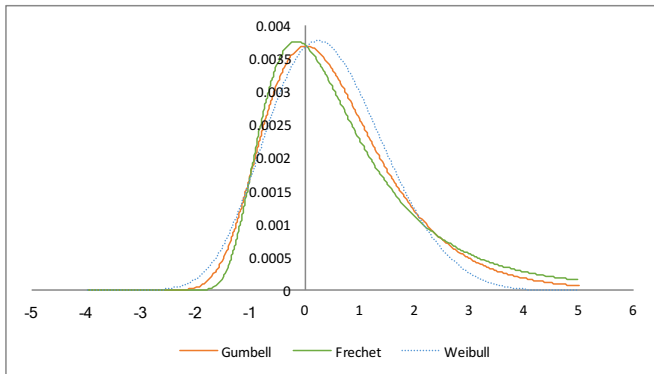
Extreme Value Theory: Two Frameworks

- There are two broad approaches to parameter estimation under extreme value theory: the block maximum framework and the peaks over threshold (POT) framework
- The block maximum framework involves separating the observations into blocks and taking the maximum from each block as constituting the extreme data points
- The POT framework specifies an arbitrary high threshold and any observed value of the series exceeding this is defined as being an extreme data point
- In each case there are three classes of distributions which normalised versions of these extremes could follow

Extreme Value Theory: Two Frameworks

- For the block maximum they are called the Weibull, Gumbel, and Frechét distributions while the corresponding distributions under the POT approach are the ordinary Pareto, exponential and beta distributions respectively.

PDFs for the Weibull, Gumbel and Frechét Distributions



The Peaks Over Threshold Approach

- The block maxima approach is rarely used since splitting the data into blocks is inefficient unless the extremes happen to be evenly spaced across the blocks
- To implement the peaks over threshold approach, we specify a threshold U and calculate the values of the data points minus the threshold:

$$\tilde{y}_t = y_t - U | y_t > U$$

- As the threshold tends to infinity, a normalised version of $\tilde{y}_t$ tends to the generalised Pareto distribution (GPD). The cdf of the GPD can be written:

$$G_{\xi, \sigma}(\tilde{y}_t) = \begin{cases} 1 - (1 + \xi \tilde{y}_t / \sigma)^{-1/\xi} & \text{if } \xi \neq 0 \\ 1 - \exp[-\tilde{y}_t / \sigma] & \text{if } \xi = 0 \end{cases}$$

where ξ is the shape parameter and σ is the scale parameter

The Peaks Over Threshold Approach

- The key parameter is ξ , sometimes also known as the tail index, measuring how rapidly the tails decay.

The Peaks Over Threshold Approach 2

- When $\xi > 0$, this corresponds to the fat-tailed case
- The tail index is the inverse of the number of degrees of freedom in a t distribution so $\xi = 1/\nu$
- Typical estimates of ν are of the order 4–6, suggesting plausible value of ξ would be 0.1–0.2
- The choice of the threshold U is tricky and involves a trade-off
- If the threshold is too big in absolute value (too far into the tail), the number of points classified as extremes will be too small leading to high standard errors for the parameter estimates (ξ and σ)

The Peaks Over Threshold Approach 2

- If the threshold is too small in absolute value (too near the centre of the distribution), many points will be classified as extremes when they are not, leading to biased parameter estimates
- One way to resolve this trade-off is to estimate the parameters for increasingly large values of U until they become stable.

Parameter Estimation for Extreme Value Distributions

- The parameters can be estimated by maximum likelihood
- The pdf for the value of y_t over the threshold (for $\xi \neq 0$) is given by:

$$g_{\xi,\sigma}(\tilde{y}_t) = \frac{1}{\sigma} \left(1 + \frac{\xi \tilde{y}_t}{\sigma} \right)^{-(\frac{1}{\xi}+1)}$$

- Then we can form the joint density for all observations over the threshold, N_U :

$$LF(\xi, \sigma, \tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_{N_U}) = \prod_{i=1}^{N_U} g_{\xi,\sigma}(\tilde{y}_t) = \prod_{i=1}^{N_U} \frac{1}{\sigma} \left(1 + \frac{\xi \tilde{y}_t}{\sigma} \right)^{-(\frac{1}{\xi}+1)}$$

Parameter Estimation for Extreme Value Distributions

- The log likelihood would then be given by taking the natural log of this and rearranging:

$$LLF(\xi, \sigma, \tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_{N_U}) = -N_U \ln(\sigma) - \left(\frac{1}{\xi} + 1\right) \sum_{i=1}^{N_U} \ln \left(1 + \frac{\xi \tilde{y}_i}{\sigma}\right)$$

Parameter Estimation for Extreme Value Distributions 2

- Provided that $\xi > -0.5$, maximum likelihood estimators are consistent and asymptotically normal
- But no analytical solutions exist and thus a numerical search procedure is required
- Instead, it is common to use a non-parametric procedure to estimate the shape parameter directly from the data
- The simplest approach is the Hill (1975) estimator

Parameter Estimation for Extreme Value Distributions 2

- If we order the exceedences over the threshold from the largest to the smallest, $\tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_T$, the Hill estimator of ξ is given by

$$\hat{\xi} = \frac{1}{k-1} \sum_{i=1}^{k-1} [\ln(\tilde{y}_{(i)}) - \ln(\tilde{y}_{(k)})]$$

where k is an integer to be selected equal to the number of observations in the tail.

Parameter Estimation for Extreme Value Distributions 3

- Two further non-parametric estimators of ξ are due to Pickands (1975)

$$\hat{\xi} = \frac{1}{\ln(2)} \ln \left(\frac{\tilde{y}_{(k)} - \tilde{y}_{(2k)}}{\tilde{y}_{(2k)} - \tilde{y}_{(4k)}} \right)$$

and due to De Haan and Resnick (1980)

$$\hat{\xi} = \frac{\ln(\tilde{y}_{(1)}) - \ln(\tilde{y}_{(k)})}{\ln(k)}$$

- If the shape parameter is estimated using one of the above non-parametric techniques, we still need to estimate the scale parameter, σ

Parameter Estimation for Extreme Value Distributions 3

- This can be achieved by plugging the given value of ξ into the LLF and then estimating σ as the only free parameter using maximum likelihood.

Introduction to Value at Risk

- Value at Risk (VaR) is a method for measuring the financial risk inherent in a portfolio or securities position
- It can be defined as the loss in financial terms that is expected to occur over a given horizon with a given degree of confidence
- Extreme value distributions can be effective for calculating VaR as they estimate the probability of extreme events more accurately than approaches not allowing for fat tails
- The delta-normal model for calculating VaR simply takes the standard deviation of the portfolio returns data, σ , and multiplies it by the relevant quantile from the normal distribution, Z_α at the α significance level

$$VaR_{normal} = \sigma Z_\alpha$$

Introduction to Value at Risk

- A second straightforward approach to calculating VaR is to use *historical simulation*, which involves sorting the actual historical portfolio returns and selecting the appropriate quantile from the empirical distribution of ordered returns.

Calculating Value at Risk using EVT

- Assume that the peaks over threshold approach is used and that the parameters are ξ and σ . VaR is calculated as follows, with a significance level of α and where N and N_U are the total number of data points and the number exceeding the threshold U respectively

$$VaR = U + \frac{\hat{\sigma}}{\hat{\xi}} \left[\left(\frac{N}{N_U} \alpha \right)^{-\hat{\xi}} - 1 \right]$$

- If the block maximum approach is used, VaR can be calculated using the following formula where m is the block length

$$VaR = \hat{\mu} + \frac{\hat{\sigma}}{\hat{\xi}} \left[1 - (-m \ln(\alpha))^{-\hat{\xi}} \right]$$

Calculating Value at Risk using EVT

- EVT can be extended to the multivariate case to measure common dependence and spillovers between extreme events in time-series.